

Rapid Microstructure Prediction in Laser Powder Bed Fusion Using Bayesian Updating of Eagar–Tsai Model

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Core Thesis

Composition



- A 5-element affine projection represents the accessible composition simplex.
- Interior coloring encodes local element content across the design space.
- Bayesian updates narrow the search to feasible LPBF compositions.

Solidification Condition

- Process parameters determine thermal gradients, cooling rates, and melt-pool shape.
- The Eagar–Tsai model provides fast temperature-field priors across the parameter space.
- These fields define the local solidification path for each candidate build condition.

Solidified Microstructure

- Microstructure emerges from the coupling of chemistry and thermal history.
- Predicted G/R conditions map to expected grain morphology and phase selection.
- The workflow targets rapid microstructure prediction from composition to final structure.



Motivation and Constraints

- LPBF alloy design spans large composition–processing spaces.
- Resulting microstructures depend on thermal gradients, G , and solidification rates, R .
- Fast thermal models are needed for high-throughput screening before expensive simulations or experiments.

Key constraint

The Eagar–Tsai model is fast enough for screening, but its simplified physics limits fidelity:

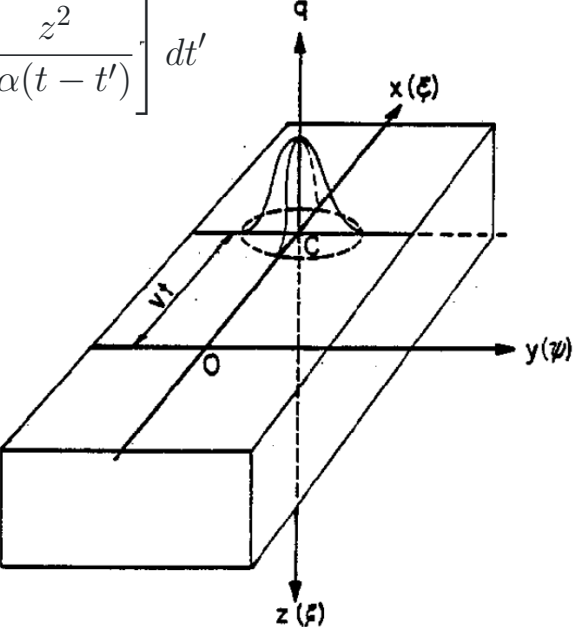
- temperature-invariant properties,
- analytical melt-pool assumptions,
- reduced accuracy relative to FEM thermal simulations.



Eagar-Tsai Model Foundation

Original Eagar-Tsai Paper

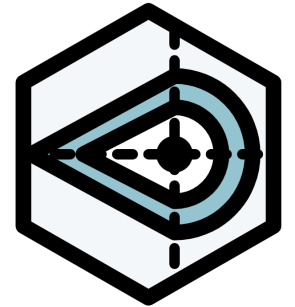
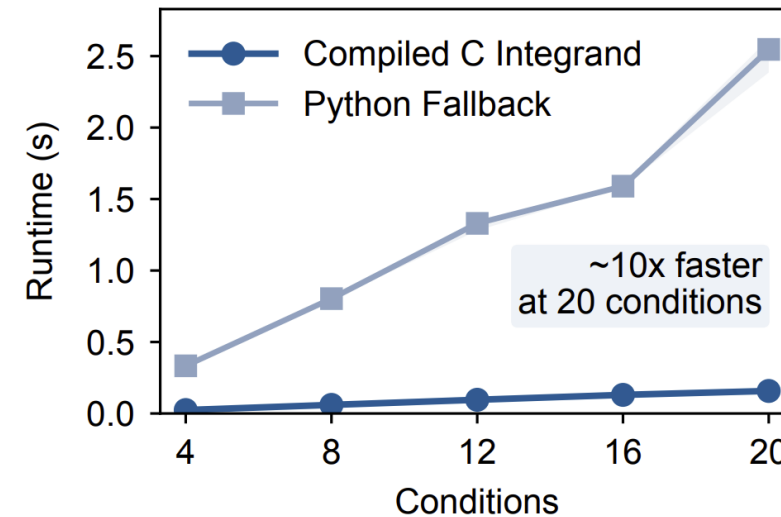
Transient Gaussian moving heat source

$$\begin{aligned} T - T_0 &= \frac{1}{2} \int_0^t dT_i \\ &= \frac{q}{\pi \rho C_p \sqrt{4\pi\alpha}} \int_0^t \frac{(t-t')^{-1/2}}{2\alpha(t-t') + \sigma^2} \\ &\quad \times \exp \left[-\frac{(x-vt')^2 + y^2}{4\alpha(t-t') + 2\sigma^2} - \frac{z^2}{4\alpha(t-t')} \right] dt' \end{aligned}$$


Eagar, T. W., & Tsai, N. S. (1983). Temperature fields produced by traveling distributed heat sources. *Welding Journal*, 62(12), 346-355.

Refactored Eagar-Tsai Model

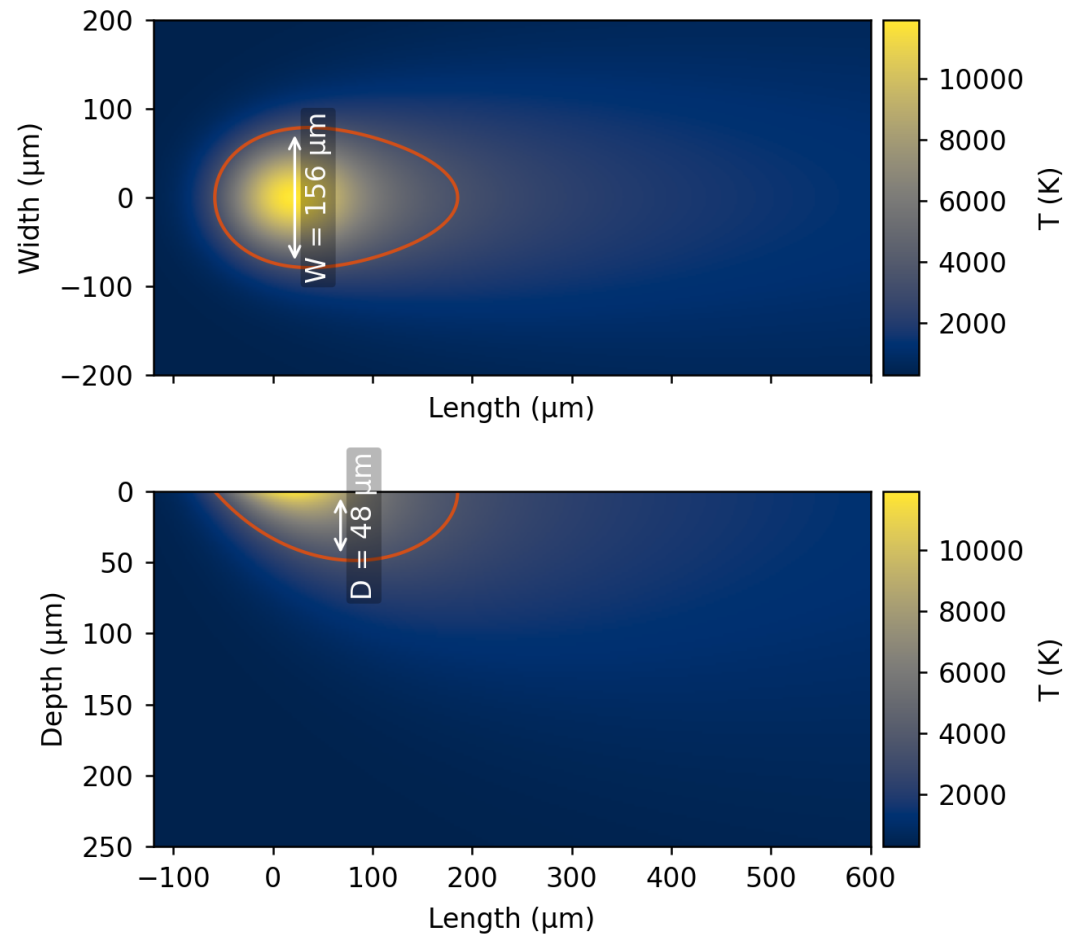
Python library implementation with compiled-C speedup



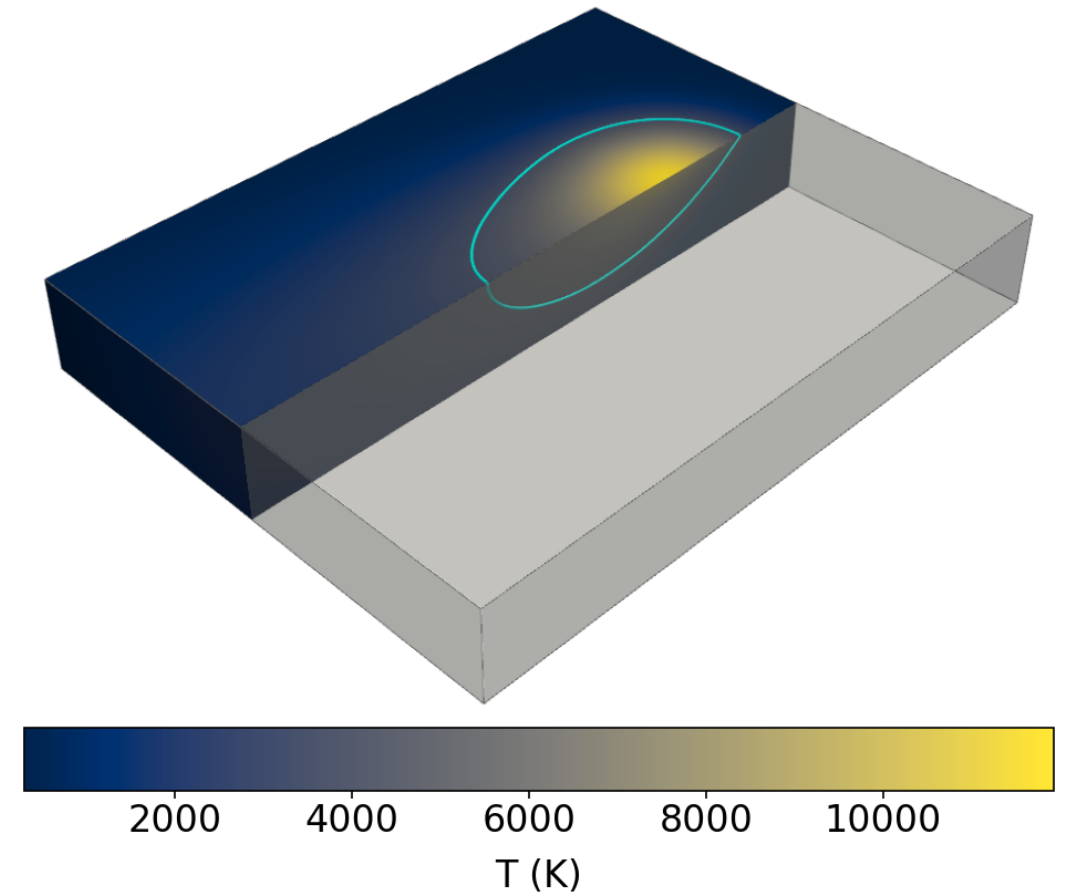
- Separates melt-pool prediction, printability maps, 2D heatmap generation, and 3D temperature-field export into reusable steps.
- Compiled C integrand removes Python overhead inside QUADPACK for faster parameter sweeps.



ET Thermal Cross Sections

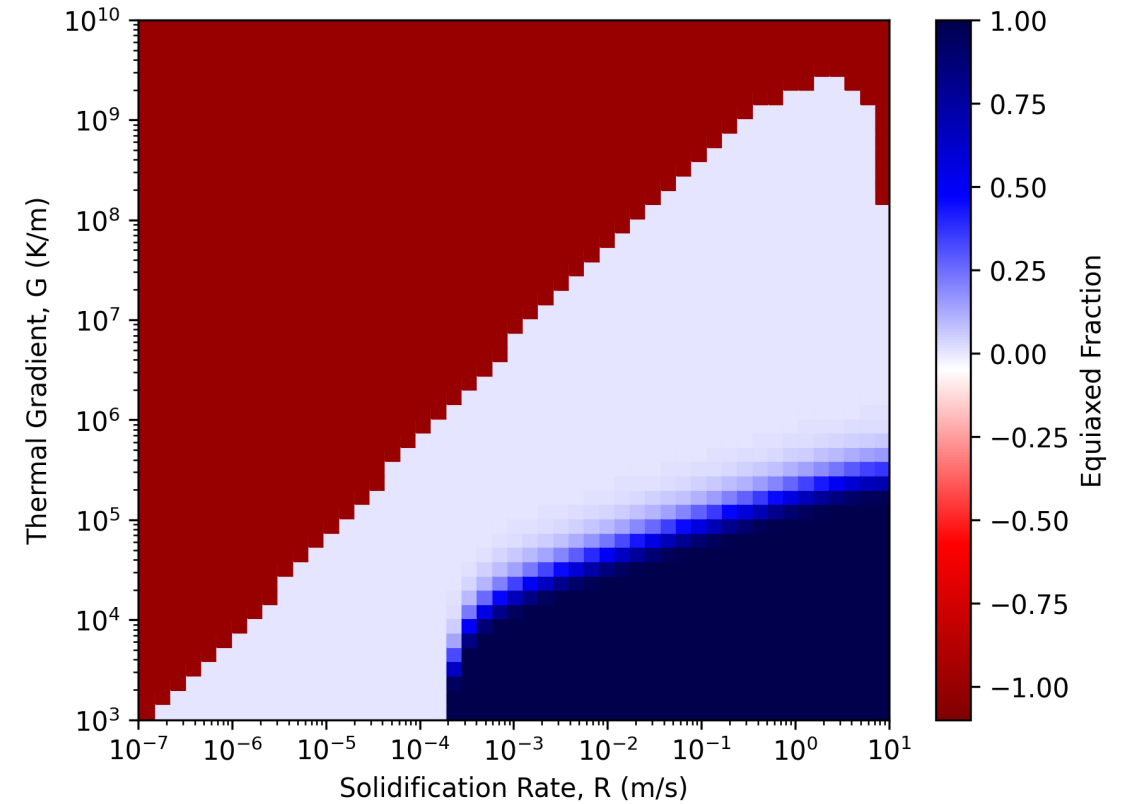
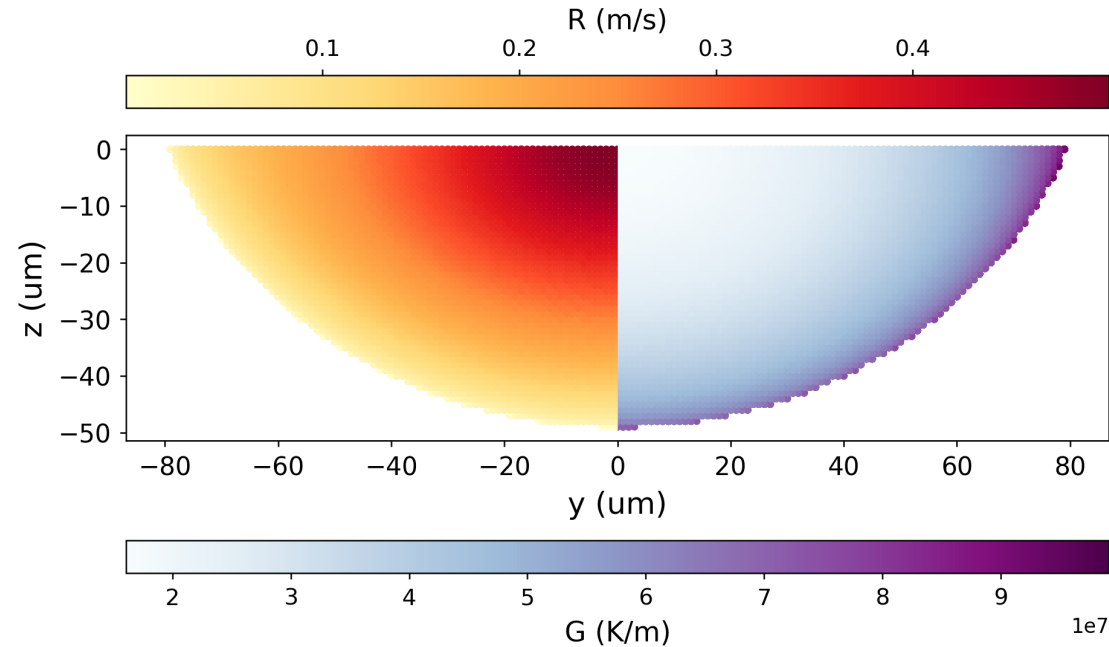


Temperature heatmap for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250 \text{ W}$ and $V = 0.5 \text{ m/s}$. Top: $z = 0$ surface. Bottom: $y = 0$ cross section.

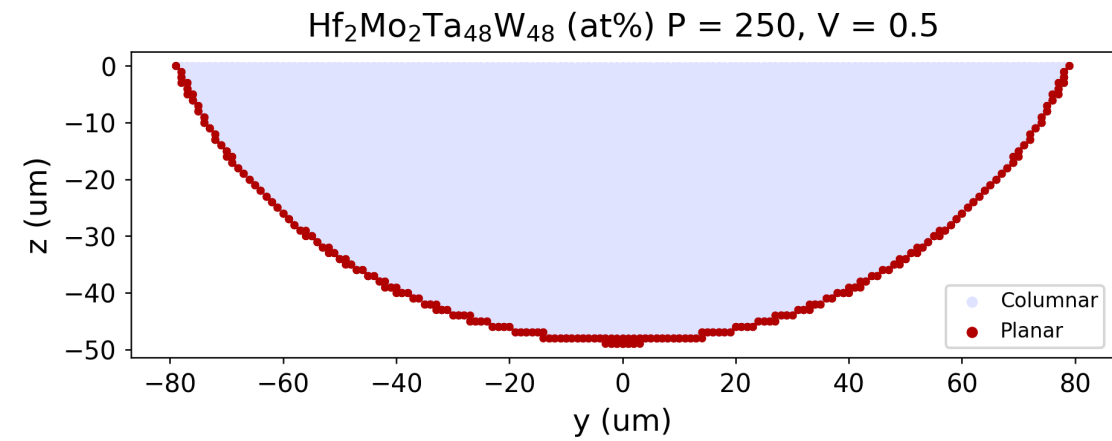
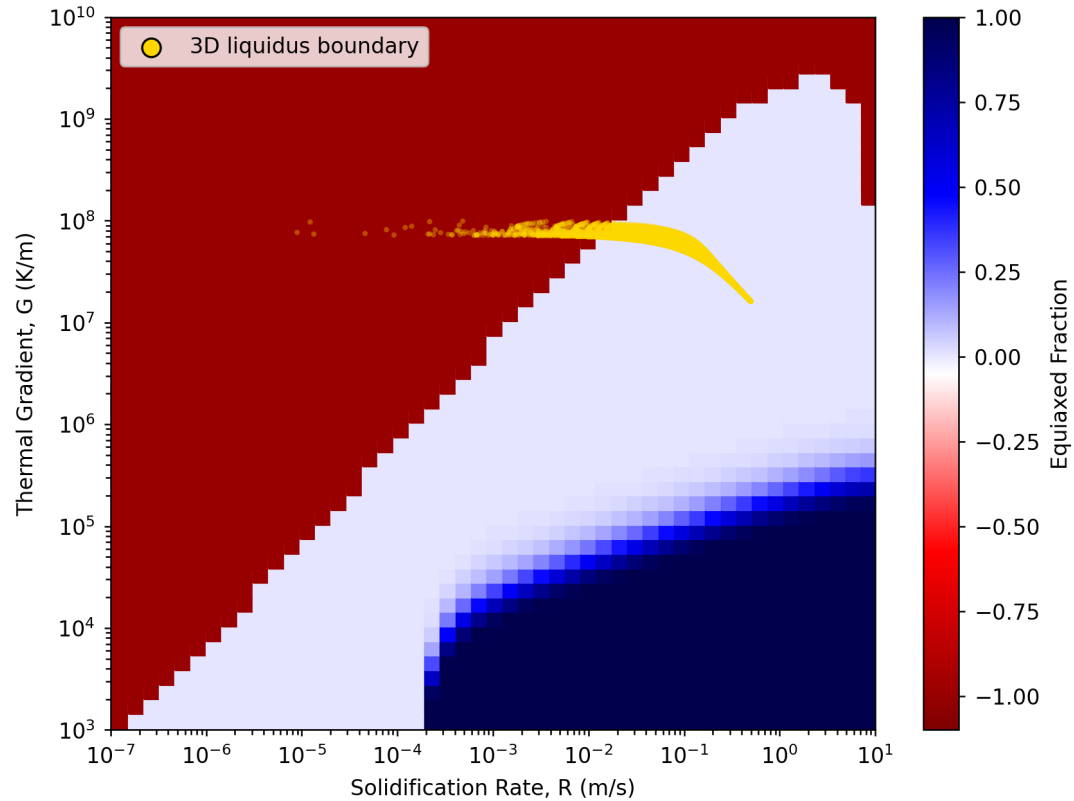


3D view of the ET melt pool for the same alloy and process condition.

ET Thermal Gradient and Solidification Rate



ET Microstructure Classification

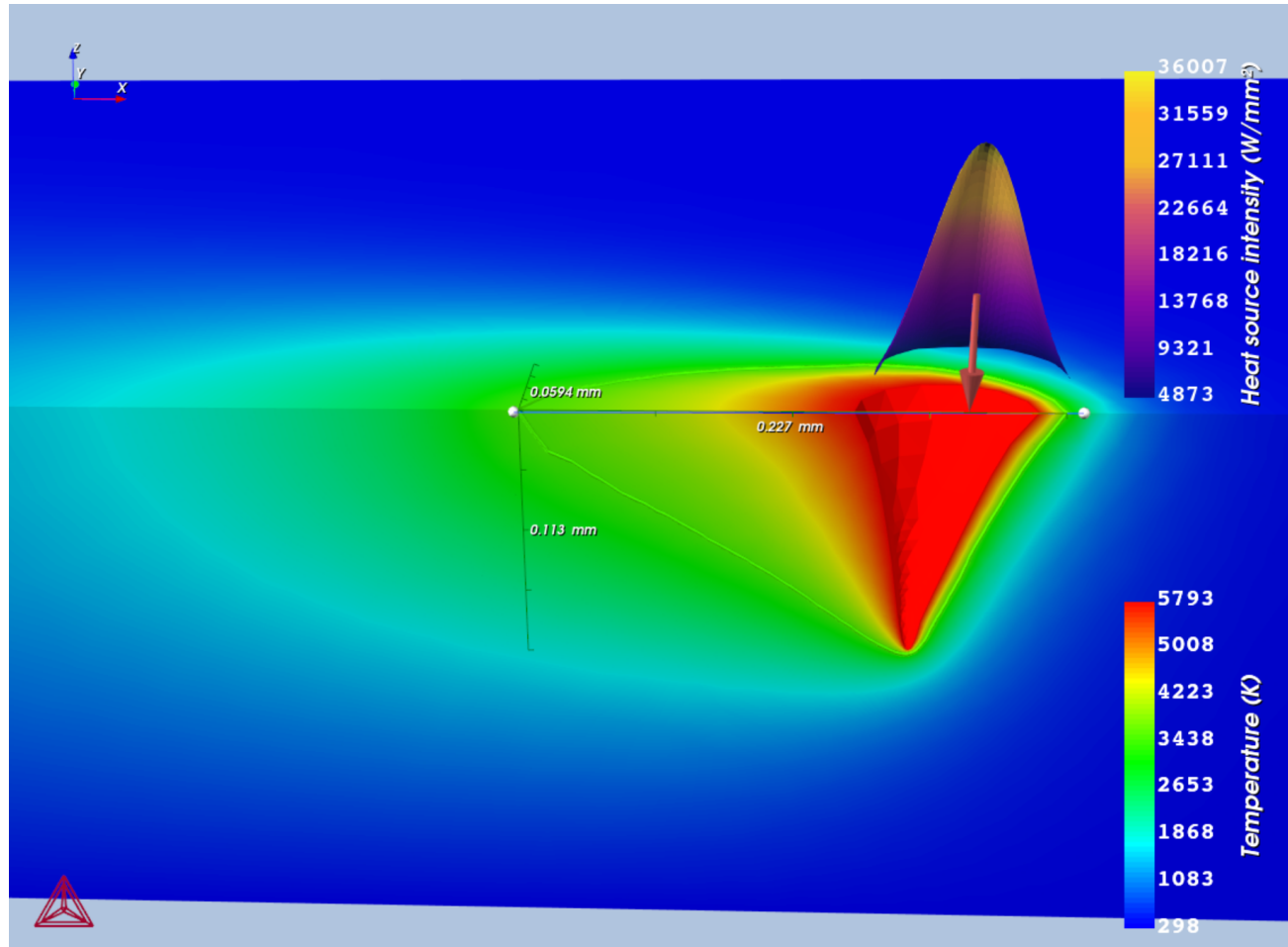


GR Map calculated with Thermo-Calc CET Property Model with overlaid GR liquidus points for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at%), $P = 250$ W and $V = 0.5$ m/s.

Projected microstructure classification along the solidification front onto a yz slice for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at%), $P = 250$ W and $V = 0.5$ m/s.



Thermo-Calc AM Module



Steady-state TCAM simulation for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

Higher-Fidelity Thermal Reference

- Composition-specific, temperature-dependent properties from TCHEA8 and Scheil calculations.
- Provides Marangoni-flow and keyholing modes, with evaporation, radiation, and convection.
- Matched composition, power, velocity, and beam dimensions enable direct comparison with Eagar-Tsai.

Outputs Used in This Work

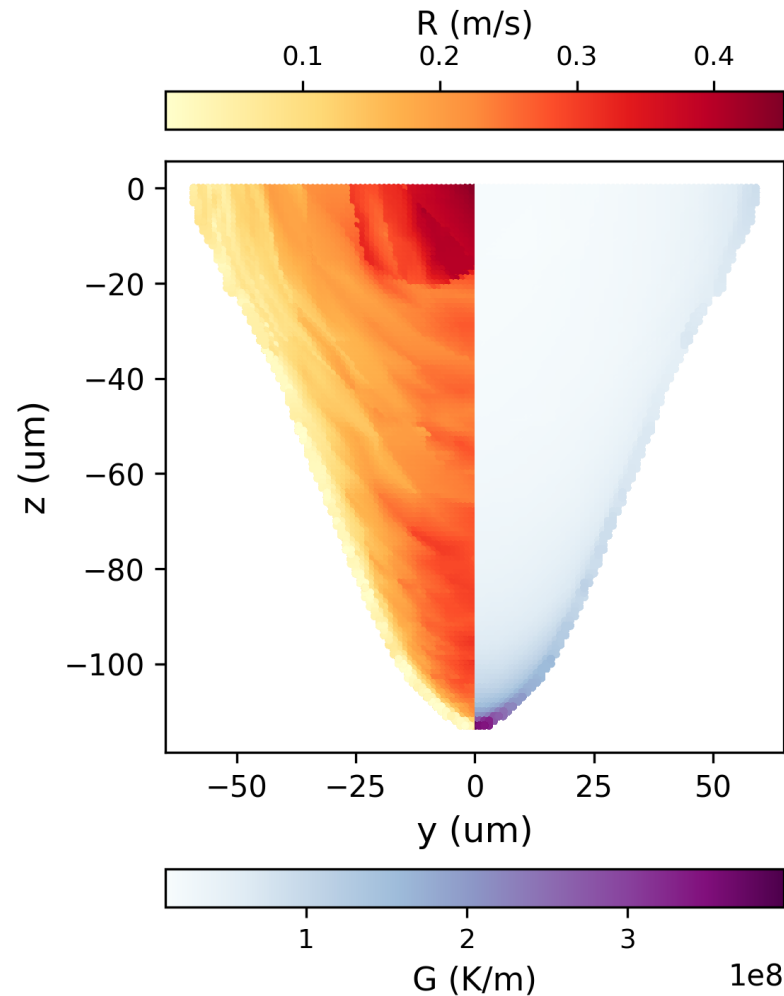
- 3D temperature field and melt-pool geometry
- G and R from downstream post-processing

Role in Bayesian Updating

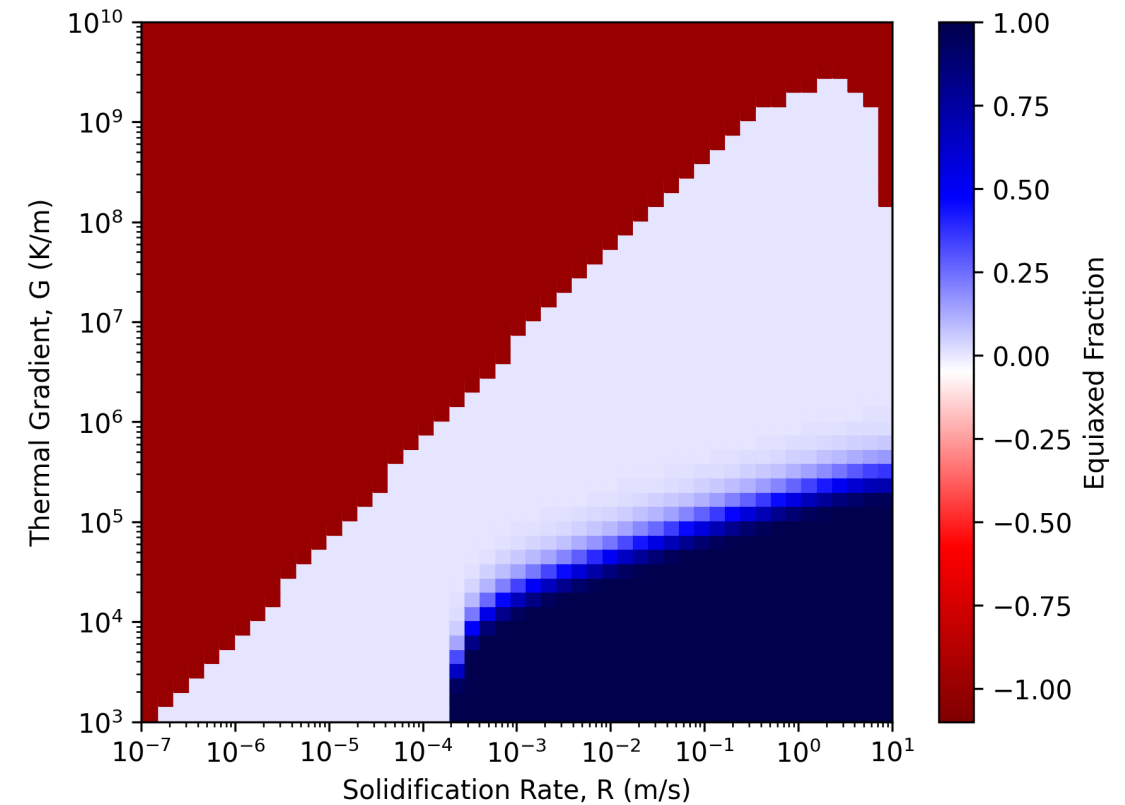
$$T_{\text{corrected}} = T_{\text{ET}} + (\widehat{T_{\text{TCAM}}} - T_{\text{ET}})$$

TCAM supplies sparse reference data used to learn and correct the ET discrepancy.

TCAM Thermal Gradient and Solidification Rate



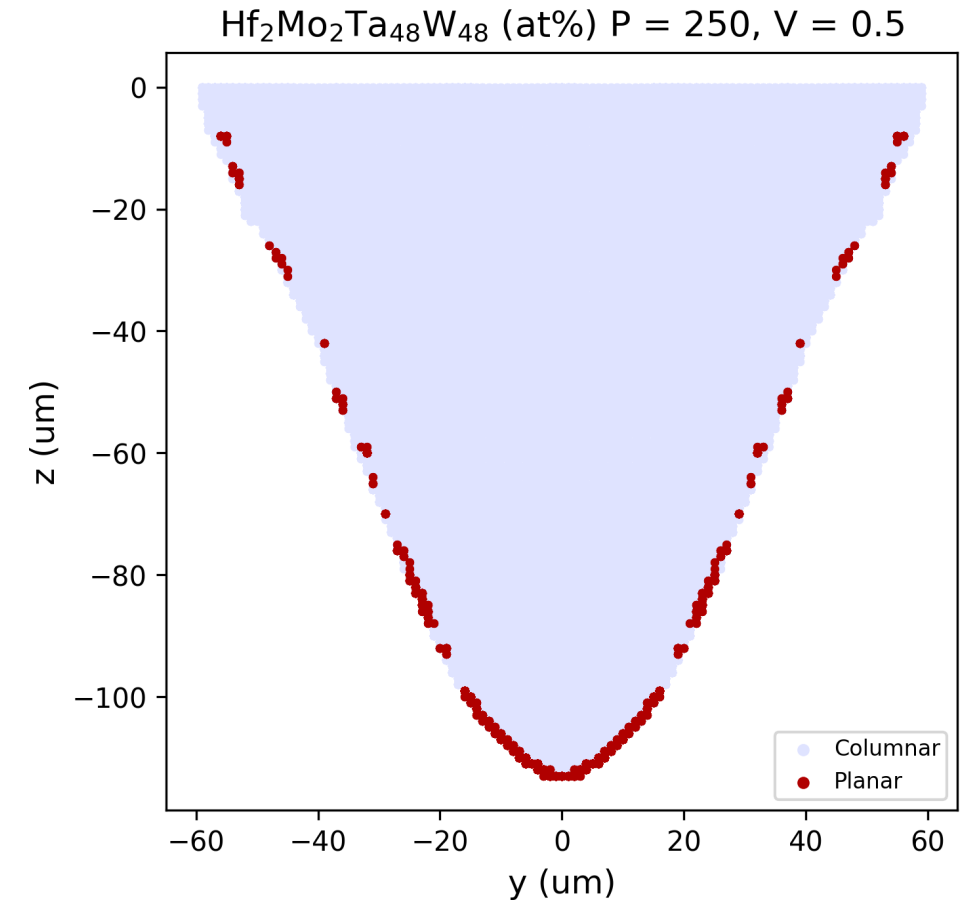
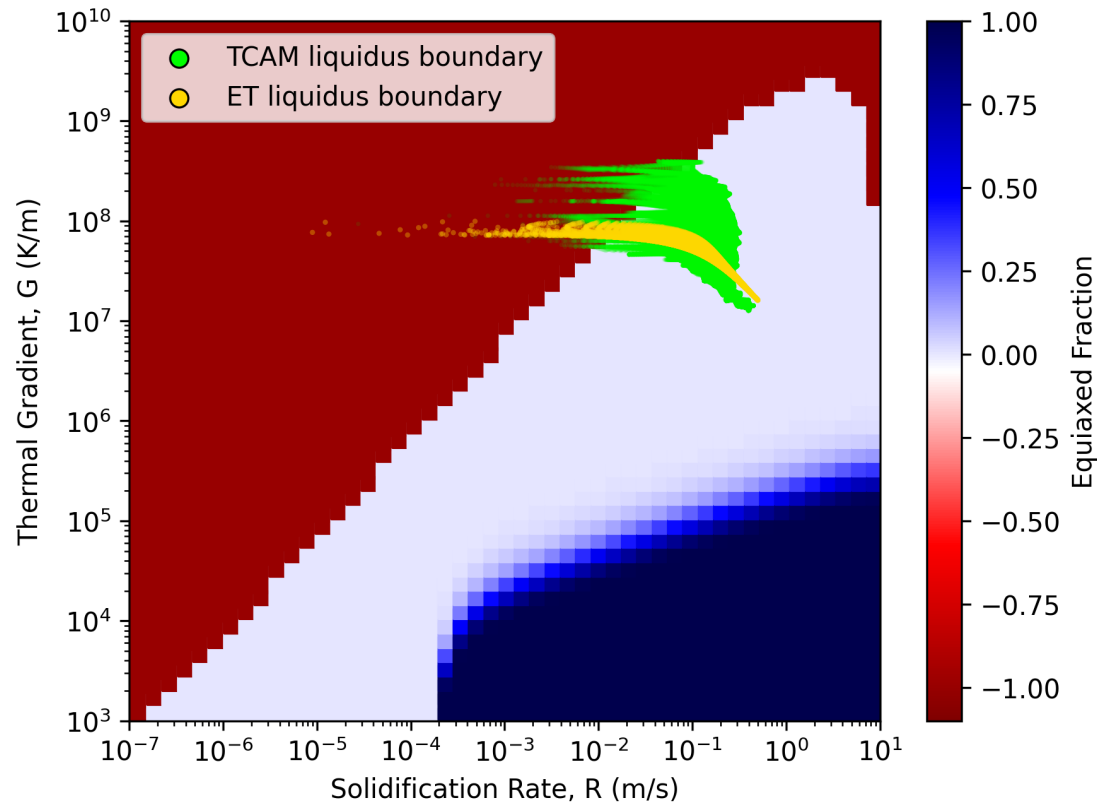
TCAM projection of G and R along the solidification front onto a yz slice for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



GR Map calculated with the Thermo-Calc CET Property Model for $\text{Hf}_2\text{Mo}_2\text{Ta}_{48}\text{W}_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



TCAM Microstructure Classification



Closing

Takeaway

End with the decision, ask, or technical conclusion you want the audience to remember.

- **Deliverable:** Name the concrete output.
- **Primary risk:** Name the hardest unresolved issue.
- **Immediate next step:** Name the action that starts now.

